

UNIT THREE

SET AND VENN DIAGRAM

GAC 135

YAO ELIKEM AYEKPLE (PhD)

KNUST

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Set Theory

Set: A collection of objects. (the objects are unordered)

Each object is an **element** of this set.

The objects that make up a set are called its **elements** (or **members**)

For example:

- The **elements** of a football team are the **players**.
- The **elements** of alphabets are **letters**.

Note:

Customarily, capital (upper) letters (such as A,B,C,S,Y) are use to designate sets and lower case letters (for example, a,b,c,s,x,y) to represent elements of sets.

Describing a set

To write a set:

- 1 $\{a, b, c\}$ explicitly listing all its elements. This is a **finite set**.
- 2 $\{x \mid x \text{ satisfies some property}\}$ This can be a **finite or infinite set**.

Set with special symbols:

- $\mathbb{N}$ = the set of natural numbers: $\{1, 2, 3, \dots\}$ (Sometimes 0 is included.)
- $\mathbb{Z}$ = the set of integers: $\{\dots, -2, -1, 0, 1, 2, \dots\}$
- $\mathbb{Q}$ = the set of rational numbers: $\{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\}$
- $\mathbb{R}$ = the set of real numbers: $\{x \mid x \in \mathbb{R}\}$ or $(-\infty, \infty)$
- $\mathbb{C}$ = the set of complex numbers: $a, b \in \mathbb{R} \mid a + bi$ where $i = \sqrt{-1}$

Terminology (Vocabulary) and Notation

- There is only one set that contains no elements, and it is called the **empty set** (or sometimes the **null set** or **void set**).
It is denoted by $\emptyset$ or $\{\}$, but not $\{\emptyset\}$.
- If a is an element of set A , it is written as: $a \in A$.
- If a is not an element of set A , it is written as: $a \notin A$.

- A set A is called a **subset** of a set B if every element of A also belongs to B . If A is a subset of B , then we write: $A \subseteq B$ or $B \supseteq A$ (read as “ A is contained in B ” or “ B contains A ”, respectively). Logically speaking, $A \subseteq B$ means $(x \in A) \rightarrow (x \in B)$ is true for every x is true for every x

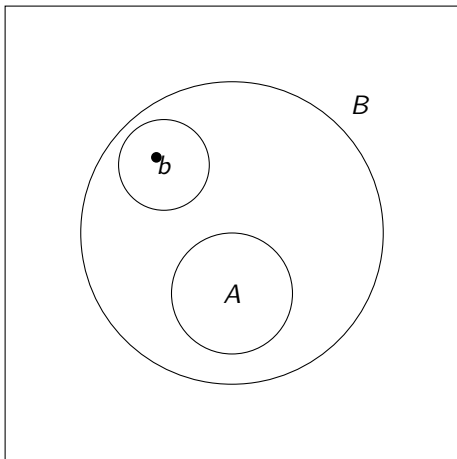
Definition

Let A and B be sets. We say that A is a **subset** of B (or A is **contained in** B) if every element of A is an element of B , and we denote this by writing $A \subseteq B$. (Occasionally, we write $B \supseteq A$ instead of $A \subseteq B$.)

If A is a subset of B

- A is a proper subset of B if
 - 1 $A \subseteq B$, and
 - 2 There exists $b \in B$ such that $b \notin A$.

This is written as $A \subset B$.
Logically speaking



Every set is a subset and super set of itself. If A is not a subset of B , we write $A \not\subseteq B$

- The empty set $\emptyset$ is a subset of every set.

- Example 0.15

Let $S = \{1, \{2\}, \{1, 2\}\}$

- Determine which of the following are elements of S :

$1, \{1\}, 2, \{2\}, \{1, 2\}, \{\{1, 2\}\}.$

- Determine which of the following are subset of S :

$\{1\}, \{2\}, \{1, 2\}, \{\{1\}, 2, \}, \{1 \{2, \}\}, \{\{1\}, \{2\}, \}, \{\{1, 2\}\}$

- Two sets A and B are **equal** if and only if every member of A is a member of B and every member of B is a member of A .

We express this by writing $A = B$.

Logically speaking, $A = B$ means $(x \in A) \equiv (x \in B)$ or the biconditional statement $(x \in A) \Leftrightarrow (x \in B)$ is true for all x .

Notes:

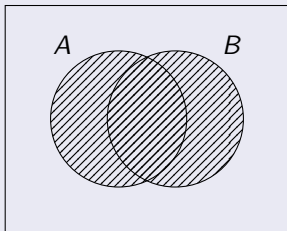
- ① The order of appearance of a set is of no consequence.
For example, the set $\{a, b, c\}$ is same as $\{b, c, a\}$ or $\{c, b, a\}$, etc.
- ② We always write each element of a set only once. For example, $\{a, a, b\}$ is not a proper way of writing a set and it should be written as $\{a, b\}$.

Algebra of Sets

Ⓐ Set Operations:

Let A , B and S be sets in the universal set (Sample Space) S ,

- ① The **union** of A and B , denoted $A \cup B$, is the set containing all elements in either A or B or both.
- ② $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
 $x \in A \cup B \iff (x \in A) \vee (x \in B)$



Example

Let $A = \{1, 3, 5\}$ and $B = \{1, 2, 3\}$.

Find $A \cup B$.

- Let $A_1, A_2, \dots, A_n$ be a finite number of sets. The union is:

$$\bigcup_{i=1}^n A_i = A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n$$

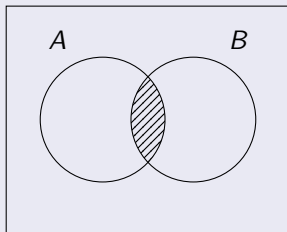
- Same way, if it is infinite number of sets,

$$\bigcup_{i=1}^{\infty} A_i = A_1 \cup A_2 \cup A_3 \cup \dots$$

- ② The **Intersection** of A and B , denoted $A \cap B$, is the set containing all elements in both A and B .

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

$$x \in A \cap B \text{ iff } (x \in A \wedge x \in B)$$



Example

Let $A = \{1, 3, 5\}$ and $B = \{1, 2, 3\}$.

Find $A \cap B$.

Let $A_1, A_2, \dots, A_n$ be finite number of sets, the intersection of sets is,

$$\bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n$$

Same way, if it is infinite number of sets, then the intersections is given as as

$$\bigcap_{i=1}^{\infty} A_i = A_1 \cap A_2 \cap A_3 \cap \dots$$

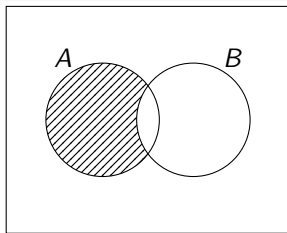
Example

Definition: Two sets are called **disjoint** if their intersection is the empty set

- ④ The **difference** of B in A , denoted $A \setminus B$ or $A - B$, is the set containing elements in A but not in B .

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}$$

$$x \in A \setminus B \text{ iff } (x \in A) \wedge \sim (x \in B).$$



Example

Let $A = \{1, 2, 5\}$ and $B = \{1, 2, 3\}$.

Find $A - B$ and $B - A$.

- 4 The **complement** of set A , denoted $\bar{A}$, A' or A^c , is the set containing all elements in S but not in A :

$$A' = \{x : x \in S \text{ and } x \notin A\}$$

$$x \in A' \text{ iff } (x \in S) \wedge \sim (x \in A).$$

- 5 Two sets A and B are called **disjoint** or **mutually exclusive** if they contain no common element, that is, if $A \cap B = \emptyset$.
- 6 Every subset of S is an event, including S and the null set $\emptyset$.
Then:

S = the certain event

$\emptyset$ = the impossible event

- If A and B are events in S , then

A' = the event that A did not occur

$A \cup B$ = the event that either A or B or both occurred.

$A \cap B$ = the event that both A and B occurred.

- Similarly, if $A_1, A_2, \dots, A_n$ are a sequence of events in S , then
 - $\bigcup_{i=1}^n A_i$ = the event that at least one of the A_i occurred.
 - $\bigcap_{i=1}^n A_i$ = the event that all of the A_i occurred.

Example 3.1

Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6\}$ be subsets of the universal set $S = \{1, 2, 3, 4, 5, 6\}$.

Find $A \cup B$, $A \cap B$, $B \setminus A$, $U \setminus B$.

Definition

If for each element j in a nonempty set J there corresponds a set A_j , then

$$\mathcal{A} = \{A_j : j \in J\}$$

is called an **indexed family** of sets with J as the index set.

- The union of all the sets $\mathcal{A}$ is defined by

$$\bigcup_{j \in J} A_j = \{x : x \in A_j \text{ for some } j \in J\}$$

- The intersection of all the sets $\mathcal{A}$ is defined by

$$\bigcap_{j \in J} A_j = \{x : x \in A_j \text{ for all } j \in J\}$$

- If $J = \{1, 2, \dots, n\}$, we may write $A_1 \cup A_2 \cup \dots \cup A_n = \bigcup_{j=1}^n A_j$ or $\bigcup_{j=1}^n A_j$ and if $J = \mathbb{N}$, the common notation is

$$\bigcup_{j=1}^{\infty} A_j \quad \text{or} \quad \bigcup_{j=1}^{\infty} A_j$$

Example 3.2

- 1 For each $k \in \mathbb{N}$, let $A_k = [0, 2 - \frac{1}{k}]$. Find $\bigcup_{k=1}^{\infty} A_k$.
- 2 Let $S = \{x \in \mathbb{R} : x > 0\}$. For each $x \in S$, let $A_x = (-\frac{1}{x}, \frac{1}{x})$. Find $\bigcap_{x \in S} A_x$.

Exercises

- 1 Mark each statement True or False. Justify each answer
- a If $A \subseteq B$ and $A \neq B$, then A is called a **proper subset** of B .
 - b The symbol $\mathbb{N}$ is used to denote the set of all integers.
 - c A slash (/) through a symbol means "not."
 - d The empty set is a subset of every set.
- 2 Mark each statement True or False. Justify each answer
- a If $A \cap B = \emptyset$, then either $A = \emptyset$ or $B = \emptyset$.
 - b If $x \in A \cup B$, then $x \in A$ or $x \in B$.
 - c If $x \in A \setminus B$, then $x \in A$ or $x \notin B$.
 - d In proving $S \subseteq T$, one should avoid beginning with "Let $x \in S$," because this assumes that S is nonempty.
- 3 Let $A = \{2, 4, 6, 8\}$, $B = \{6, 7, 8, 9\}$, and $C = \{2, 8\}$. Which of the following statements are true?
- | | |
|------------------------------------|---------------------------------------|
| a $\{8, 7\} \subseteq B$ | e $\emptyset \in B$ |
| b $\{7\} \subseteq B \cap C$ | f $A \cap B \cap C = 8$ |
| c $(A \setminus B) \cap C = \{2\}$ | g $B \setminus A = \{2, 4\}$ |
| d $C \setminus A = \emptyset$ | h $(B \cup C) \setminus A = \{7, 9\}$ |

- 4 Let $A = \{2, 4, 6, 8\}$, $B = \{6, 8, 10\}$, and $C = \{5, 6, 7, 8\}$. Find the following sets

a $A \cap B$

b $A \cup B$

c $A \setminus B$

d $B \cap C$

e $B \setminus C$

f $(B \cup C) \setminus A$

g $(A \cap B) \setminus C$

h $C \setminus (A \cup B)$

- 5 Use Venn diagrams with three overlapping circles to illustrate each identity.

a $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

b $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$

- 6 Let A and B be subsets of a universal set S . Simplify each of the following expressions.

a $(A \cup B) \cup (S \setminus A)$

b $(A \cap B) \cap (S \setminus A)$

c $A \cap [B \cup (S \setminus A)]$

d $A \cup [B \cap (S \setminus A)]$

e $(A \cup B) \cap [A \cup (S \setminus B)]$

f $(A \cap B) \cup [A \cap (S \setminus B)]$

- 7 Which statement(s) below would enable one to conclude that $x \in A \cap B$?
- a $x \in A$ and $x \in B$.
 - b $x \in A$ or $x \in B$.
 - c $x \in A$ and $x \notin A \setminus B$.
 - d If $x \in A$, then $x \in B$.
- 8 Which statement(s) below would enable one to conclude that $x \in A \setminus B$?
- a $x \in A$ and $x \notin B \setminus A$.
 - b $x \in A \cup B$ and $x \notin B$.
 - c $x \in A \cup B$ and $x \notin A \cap B$.
 - d $x \in A$ and $x \notin A \cap B$.
- 9 Which statement(s) below would enable one to conclude that $x \notin A \setminus B$?
- a $x \notin A \cup B$.
 - b $x \in B \setminus A$.
 - c $x \in A \cap B$.
 - d $x \in A \cup B$ and $x \notin A$.
 - e $x \in A \cup B$ and $x \notin A \cap B$.
- 10 Find $\bigcup_{B \in \mathcal{B}} B$ and $\bigcap_{B \in \mathcal{B}} B$ for each collection $\mathcal{B}$.
- a $\mathcal{B} = \{[1, 1 + \frac{1}{n}] : n \in \mathbb{N}\}$
 - b $\mathcal{B} = \{(1, 1 + \frac{1}{n}) : n \in \mathbb{N}\}$
 - c $\mathcal{B} = \{[2, x] : x \in \mathbb{R} \text{ and } x > 2\}$
 - d $\mathcal{B} = \{[0, 3], (1, 5), [2, 4]\}$
- 11 Let $\{A_j : j \in J\}$ be an indexed family of sets and let B be a set. Prove the following generalizations of Theorem 1.13.

Ordered Pairs

Definition:

The **ordered pair** (a, b) is the set whose numbers are $\{a\}$ and $\{a, b\}$. In symbols we have

$$(a, b) = \{\{a\}, \{a, b\}\}.$$

Theorem:

$$(a, b) = (c, d) \text{ iff } a = c \text{ and } b = d.$$

Example 3.3

Let $A = \{1, 2\}$. List all the ordered pairs (x, y) such that $x \in A$ and $y \in A$.

Cartesian Products

Definition:

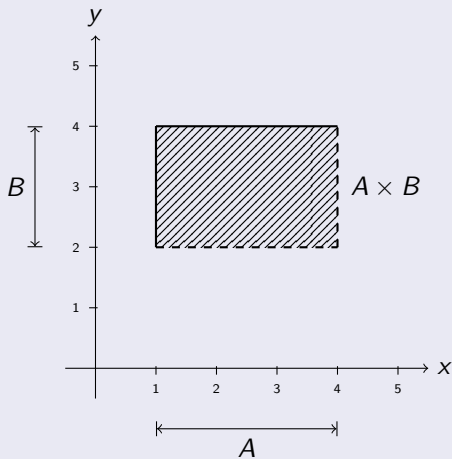
If A and B are sets, then the **Cartesian product** (or **cross product**) of A and B written $A \times B$, is the set of all ordered pairs (a, b) such that $a \in A$ and $b \in B$. In symbols,

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$$

For example, if $A = [1, 4)$ and $B = (2, 4]$, then

$$A \times B = \{(x, y) : 1 \leq x < 4 \text{ and } 2 < y \leq 4\}$$

Geometrically,



Example 3.4

Consider the following intervals:

$$A = (1, 5] \quad B = [1, 4)$$

$$C = [4, 7) \quad D = (2, 5)$$

Draw a sketch to illustrate that

$$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$$

CARDINALITY

The cardinality of a set is simply the number of unique elements it contains, that is the set's "size",

- For example: If $S = \{x \in \mathbb{R} : x^2 = 25\}$, then $S = \{-5, 5\}$ and we say S has two elements. If $T = \{2, 3, 5\}$, then T has three elements and we think of T as being "larger" than S .
- If two sets S and T are the same size, then there is a bijective function $f : S \rightarrow T$, for this function will set up a one-to-one correspondence between the elements of each set.

Definition

- Two sets S and T are called **equinumerous** (have the same cardinality), and we write $S \sim T$, if there exists a bijective function from S onto T .

$$f : S \rightarrow T$$

Cardinal Numbers

Two sets have the same cardinal number if and only if they are equinumerous.

Definition

- A set S is said to be **finite** if $S = \emptyset$ or if there exists an $n \in \mathbb{N}$ and a bijection

$$f : \{1, 2, \dots, n\} \rightarrow S.$$

- If a set is not finite, it is said to be **infinite**.
- Let $\{1, 2, \dots, n\}$ be denoted by I_n .

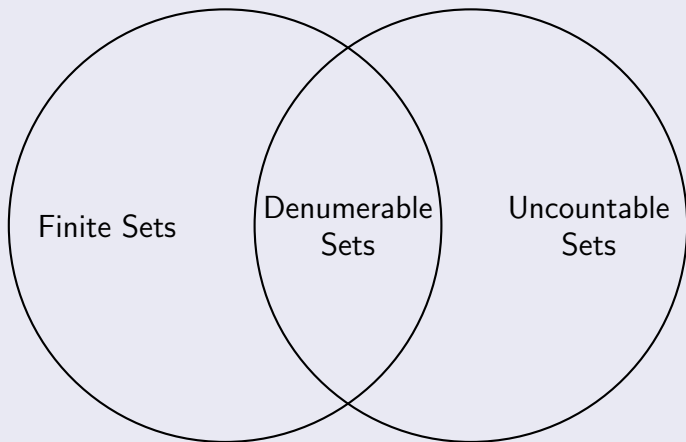
Definition

- The cardinal number of I_n is n , and if $S \sim I_n$, we say that S has n **elements**.
- The cardinal number of $\emptyset$ is taken to be 0.
- If a cardinal number is not finite, it is called transfinite.

Definition

- A set S is said to be **denumerable** if there exists a bijection $f : \mathbb{N} \rightarrow S$.
- If a set is a finite or denumerable, it is called **countable**.
- If a set is not countable, it is **uncountable**.
- The cardinal number of a denumerable set is denoted by $\aleph_0$ (aleph-nought).
- In other words, a set is denumerable iff it is equinumerous with the set of natural numbers $\mathbb{N}$.

Countable Sets Infinite Sets



Theorem:

Let S be a countable set and $T \subseteq S$. Then T is countable.

Theorem:

Let S be a non-empty set.

The following three conditions are equivalent.

- a) S is countable.
- b) There exists an injection $f : S \rightarrow \mathbb{N}$.
- c) There exists a surjection $g : \mathbb{N} \rightarrow S$.

Theorem:

The set $\mathbb{R}$ of real numbers is uncountable.

The cardinal number of $\mathbb{R}$ is denoted by $\mathfrak{c}$.

$$\text{i.e., } |\mathbb{R}| = \mathfrak{c}$$

where $\mathfrak{c}$ stands for continuum.

Note: Since $\mathbb{Q} \subseteq \mathbb{R}$, we have $\aleph_0 \leq \mathfrak{c}$.

In fact, since $\mathbb{Q}$ is countable and $\mathbb{R}$ is uncountable, we have $\aleph_0 < \mathfrak{c}$

Power Set

The **power set** (or **powerset**) of a set S is the set of all subsets of S , including the empty set and S itself. The power set of S is denoted by

$$\mathcal{P}(S) \text{ or } 2^S$$

- Any subset of $\mathcal{P}(S)$ is called a *family of sets* over S . For example; If $S = \{a, b, c\}$, then all the subsets of S are
 - $\{ \}$ or $\emptyset$
 - $\{a\}$
 - $\{b\}$
 - $\{c\}$
 - $\{a, b\}$
 - $\{a, c\}$
 - $\{b, c\}$
 - $\{a, b, c\} = S$

and hence the power set of S ,

$$\mathcal{P}(S) = \{ \{ \}, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\} \}.$$

Venn Diagram

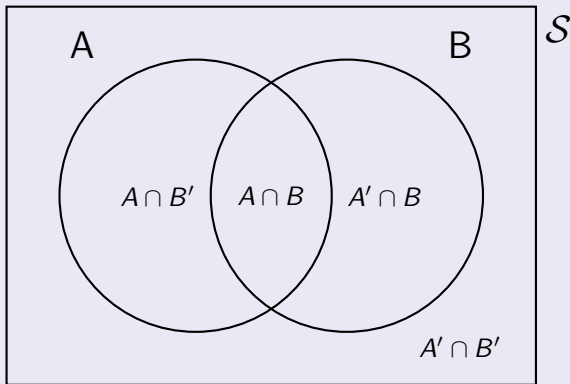
Two-set problems

If A and B are two subsets of S , then every $x \in S$ belongs to exactly one of the following four subsets of S :

$$A \cap B, \quad A \cap B', \quad A' \cap B \quad \text{and} \quad A' \cap B'.$$

The set

- $A \cap B$ consists of elements belonging to both A and B ;
- $A \cap B'$ consists of elements belonging to A that do not belong to B ;
- $A' \cap B$ consists of elements belonging to B that do not belong to A ;
- $A' \cap B'$ consists of elements that belong to neither A nor B .



Three-set problems

The set

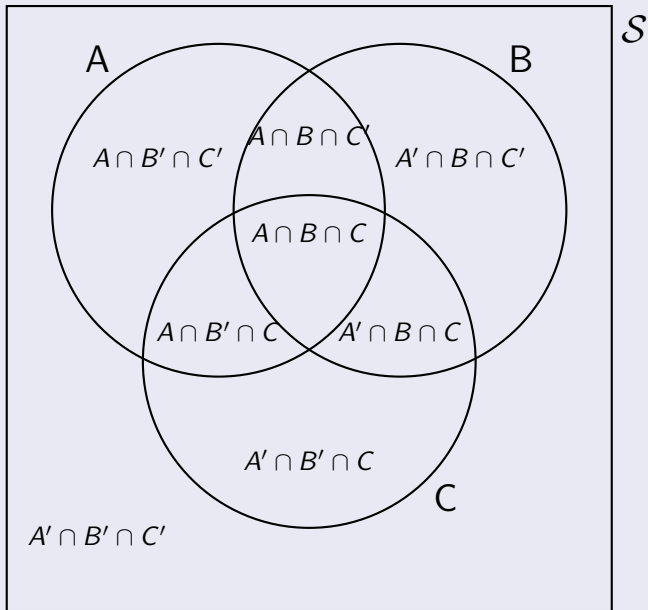
$A \cap B \cap C$ consists of elements belonging to A , B and C ;

$A \cap B' \cap C$ consists of elements belonging to A and C but not B ;

$A \cap B \cap C'$ consists of elements belonging to A and B but not C ;

$A \cap B' \cap C'$ consists of elements belonging to A but not B and C ;

$A' \cap B' \cap C'$ consists of elements that do not belong to any of the sets A , B and C .



Example 3.5

1.(a) Draw a Venn diagram of a universal set S and two intersecting finite subsets A and B . Let $n(A)$ denote the number of members in the set A . If $n(A \cap B) = x$, $n(A \cap B') = y$ and $n(B \cap A') = z$, find $n(A)$, $n(B)$, $n(A \cup B)$ in terms of x , y and z and deduce an expression for $n(A \cup B)$ in terms of $n(A)$, $n(B)$ and $n(A \cap B)$.

What does this expression reduce to when A and B are disjoint?

Determine the relationship between $n(A \cup B)$ and $n(A \cap B)$ if

- (i) $A = B$,
 - (ii) $A = A \cup B$.
- (b)** In a class of 20 boys, 16 play soccer, 12 play hockey and two are not allowed to play games. Find how many students play
- (i) soccer and hockey,
 - (ii) only hockey.

Example 3.6

(a) A recent survey of 40 students revealed that the number studying one or more of the three subjects mathematics (M), physics (P) and chemistry (C) is as follows:

Subject	Number of students
M	22
M and P	5
M and C	8
P and C	5
M , P and C	3

If the number of students who study physics as their only subject is the same as that who study chemistry as their only subject, find the number of students who study

- (i) only physics,
- (ii) only one subject.

(b) All the inhabitants of a certain country speak either French and/or English. If 60% speak French and 50% speak English, how many speak both languages?

Exercises

1. Mark each statement True or False. Justify each answer.

- a Two sets S and T are equinumerous if there exists a bijection $f : S \rightarrow T$.
- b If a set S is finite, then S is equinumerous with I_n for some $n \in \mathbb{N}$.
- c If a cardinal number is not finite, it is said to be infinite.
- d A set S is denumerable if there exists a bijection $f : \mathbb{R} \rightarrow S$.
- e Every subset of a countable set is countable.
- f Every subset of a denumerable set is denumerable.

2. Mark each statement True or False. Justify each answer.

- a A nonempty set S is countable iff there exists an injection $f : \mathbb{N} \rightarrow S$.
- b The set $\mathbb{Q}$ of rational numbers is denumerable.
- c Let S be a nonempty set. There exists an injection $f : S \rightarrow \mathbb{N}$ iff there exists a surjection $g : \mathbb{N} \rightarrow S$.

Exercise Continued

- d) The set $\mathbb{R}$ of real numbers is denumerable.
- e) $|S| \leq |T|$ means that there exists an injection $f : S \rightarrow T$
- f) The continuous hypothesis says that $\aleph_0$ is the smallest transfinite cardinal number

3. Show that the following pairs of sets S and T are equinumerous by finding a specific bijection between the sets in each pair.

- a) $S = [0, 1]$ and $T = [1, 4]$
- b) $S = [0, 1]$ and $T = [2, 7]$
- c) $S = [0, 1]$ and $T = [0, 1)$
- d) $S = [0, 1)$ and $T = (0, 1)$
- e) $S = (0, 1)$ and $T = (0, \infty)$
- f) $S = (0, 1)$ and $T = \mathbb{R}$

4. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{0, 3, 6\}$. Find:

- | | |
|---------------|---------------|
| a) $A \cup B$ | b) $A \cap B$ |
| c) $A - B$ | e) $B - A$ |

Exercise Continued

5. Let $A = \{a, b, c, d, e\}$ and $B = \{a, b, c, d, e, f, g, h\}$. Find:

a) $A \cup B$

b) $A \cap B$

b) $A - B$

c) $B - A$

6. Let $A_i = \{1, 2, 3, \dots, i\}$ for $i = 1, 2, 3, \dots$. Find:

a) $\bigcup_{i=1}^n A_i$

b) $\bigcap_{i=1}^n A_i$

7. Let $A_i = \{-i, \dots, -2, -1, 0, 1, \dots, i\}$. Find:

a) $\bigcup_{i=1}^n A_i$

b) $\bigcap_{i=1}^n A_i$

8. Let A_i be the set of all nonempty bit strings (that is, bit strings of length at least one) of length not exceeding i . Find:

a) $\bigcup_{i=1}^n A_i$

b) $\bigcap_{i=1}^n A_i$

Exercise Continued

9. Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$ if for every positive integer i ,

- a) $A_i = \{i, i+1, i+2, \dots\}$.
- b) $A_i = \{0, i\}$.
- c) $A_i = (0, i)$, that is, the set of real numbers x with $0 < x < i$.
- d) $A_i = (i, \infty)$, that is, the set of real numbers x with $x > i$.

10. A survey of the reading habits of 100 students showed that 30 read both Comics and Novels, 10 read neither Comics nor Novels and twice as many students read Comics as read Novels.

- i) Illustrate the above information in a Venn diagram
- ii) How many students read novels?
- iii) How many students read comics only?

Exercise Continued

11. In a science class of 40 students, 22 study Mathematics, 5 study Mathematics and Physics, 8 study Mathematics and Chemistry, 5 study Physics and Chemistry and 3 study all the three subjects. If the number of students who study only Physics is equal to that who study only Chemistry, illustrate the given information in a Venn diagram and find the number of students who study

- i) Only Physics
- ii) Chemistry
- iii) Only one subject